

Lecture 2.1: RAS and MYC - The Oncogenic Powerhouses

Good morning, everyone. Today we're embarking on a detailed exploration of two of the most significant oncogene families in cancer biology: RAS and MYC. These aren't merely important genes—they represent the molecular engines that drive some of the most aggressive and therapeutically challenging cancers known to medicine.

What makes RAS and MYC particularly fascinating is that they represent two fundamentally different yet complementary approaches to cellular transformation. RAS functions as a signal transduction switch that hijacks normal growth control mechanisms, while MYC acts as a transcriptional master regulator that reprograms the entire cellular machinery for uncontrolled proliferation. Together, they illustrate the sophisticated molecular strategies that cancer cells employ to achieve malignant transformation.

The RAS Family: Molecular Switches in Cancer

Let's begin our journey with the RAS family, consisting of three closely related but functionally distinct genes: KRAS, NRAS, and HRAS. The discovery of RAS genes fundamentally changed our understanding of cancer by revealing that normal cellular growth control mechanisms could be subverted by relatively simple molecular alterations.

The GTPase Switch: Molecular Architecture of Control

To understand how RAS drives cancer, we must first appreciate its role as a molecular switch. RAS proteins belong to the small GTPase superfamily, which represents one of nature's most elegant solutions to the problem of cellular signal transduction.

The fundamental principle underlying RAS function is binary switching based on nucleotide binding. When RAS binds GTP, it adopts an active conformation capable of interacting with downstream effector proteins. When it binds GDP, it assumes an inactive conformation that cannot productively engage these same effectors.

This switching mechanism is controlled by two classes of regulatory proteins. Guanine nucleotide exchange factors, or GEFs, catalyze the exchange of GDP for GTP, activating RAS. GTPase-activating proteins, or GAPs, accelerate the intrinsic GTPase activity of RAS, converting GTP to GDP and inactivating the protein.

In normal cells, this creates a tightly regulated cycle where RAS activity is precisely controlled in response to extracellular signals. Growth factors activate receptor tyrosine kinases, which recruit and activate specific GEFs, leading to RAS activation. Once the growth signal diminishes, GAPs restore RAS to its inactive state.

Structural Biology: The Molecular Basis of Switching

The three-dimensional structure of RAS proteins reveals how this elegant switching mechanism operates at the atomic level. RAS contains two critical structural elements called Switch I and Switch II regions, which undergo dramatic conformational changes depending on whether GTP or GDP is bound.

When GTP is bound, these switch regions adopt conformations that create binding surfaces for effector proteins. The gamma phosphate of GTP makes crucial contacts with specific amino acid residues, stabilizing the active conformation. When GTP is hydrolyzed to GDP, the loss of these contacts causes the switch regions to collapse into conformations that cannot bind effectors.

This structural mechanism explains why RAS is such an effective molecular switch. The conformational changes are substantial enough to completely alter protein-protein interaction surfaces, yet they're triggered by the simple chemical difference between GTP and GDP.

The crystal structures also reveal why certain amino acid positions are particularly vulnerable to oncogenic mutation. Residues at positions 12, 13, and 61 are directly involved in either GTP hydrolysis or the positioning of catalytic residues. Mutations at these positions can maintain the GTP-binding capability while impairing the hydrolysis reaction.

Oncogenic Activation: When Switches Get Stuck

The vast majority of oncogenic RAS mutations occur at just three amino acid positions: codons 12, 13, and 61. These mutations represent a perfect example of how cancer can result from the disruption of normal regulatory mechanisms.

Codon 12 mutations are the most frequent, accounting for approximately 80-85% of oncogenic RAS mutations. The wild-type glycine at this position is crucial for proper positioning of the catalytic machinery involved in GTP hydrolysis. When glycine is replaced by amino acids with bulkier side chains—such as aspartic acid, valine, or cysteine—the catalytic geometry is disrupted.

Codon 13 mutations operate through a similar mechanism but are less common, representing about 10-15% of RAS mutations. The wild-type glycine at position 13 also participates in the GTPase reaction, and its substitution impairs catalytic efficiency.

Codon 61 mutations, while less frequent, are equally significant. Position 61 contains a glutamine residue that directly participates in GTP hydrolysis by positioning a water molecule for nucleophilic attack on the gamma phosphate. Mutations at this position, particularly to histidine or leucine, severely impair the hydrolysis reaction.

The key insight is that these mutations don't prevent GTP binding or grossly disrupt protein structure. Instead, they selectively impair the protein's ability to turn itself off. The result is a constitutively active RAS protein that continuously sends proliferative signals regardless of external growth factor availability.

Signaling Networks: Multiple Pathways from One Switch

The power of RAS as an oncogene derives not just from its constitutive activation, but from its ability to simultaneously activate multiple downstream signaling pathways. RAS doesn't simply turn on proliferation—it orchestrates a complex program of cellular changes that contribute to multiple cancer hallmarks.

The RAF-MEK-ERK Pathway: The Proliferation Engine

The most extensively studied RAS effector pathway involves the sequential activation of RAF, MEK, and ERK kinases. When active RAS-GTP binds to RAF kinases, it triggers their activation through a complex mechanism involving membrane recruitment, conformational changes, and phosphorylation events.

Activated RAF then phosphorylates and activates MEK1 and MEK2, which are dual-specificity kinases that phosphorylate both threonine and tyrosine residues on their substrates. MEK, in turn, phosphorylates and activates ERK1 and ERK2.

ERK represents the business end of this pathway. Once activated, ERK translocates to the nucleus where it phosphorylates numerous transcription factors, including ELK1, FOS, and JUN. These transcription factors drive the expression of immediate early genes that promote cell cycle progression and proliferation.

The RAF-MEK-ERK pathway is particularly important for the "sustaining proliferative signaling" hallmark of cancer. Constitutively active RAS leads to persistent ERK signaling, driving cells to proliferate regardless of external growth factor signals.

The PI3K-AKT-mTOR Pathway: The Survival Network

RAS also activates the PI3K-AKT-mTOR pathway, which is crucial for cell survival and growth. RAS-GTP can directly bind to the p110 catalytic subunit of PI3K, activating its lipid kinase activity.

Activated PI3K phosphorylates phosphatidylinositol 4,5-bisphosphate to generate phosphatidylinositol 3,4,5-trisphosphate (PIP3), a lipid second messenger that recruits AKT to the plasma membrane. At the membrane, AKT is phosphorylated and activated by PDK1 and mTORC2.

Active AKT has numerous pro-survival and growth-promoting effects. It phosphorylates and inactivates pro-apoptotic proteins like BAD and FOXO transcription factors. It activates MDM2, leading to p53 degradation. It also activates mTORC1, a master regulator of protein synthesis and cellular growth.

This pathway is essential for the "resisting cell death" hallmark of cancer. RAS activation through PI3K-AKT signaling helps cancer cells survive in conditions that would normally trigger apoptosis.

The RAL-GEF Pathway: Cytoskeletal Control and Beyond

The third major RAS effector pathway involves RAL guanine nucleotide exchange factors (RAL-GEFs), which activate RAL GTPases. This pathway is less well understood but equally important for cancer development.

RAL proteins regulate various cellular processes including vesicle trafficking, cytoskeletal organization, and transcriptional control. RAL signaling contributes to several cancer hallmarks, including invasion and metastasis through its effects on cell adhesion and motility.

Clinical Significance: The KRAS Dominance

While all three RAS family members can cause cancer when mutated, KRAS stands out for its overwhelming clinical importance. KRAS mutations are found in approximately 95% of pancreatic adenocarcinomas, 50% of colorectal cancers, and 30% of lung adenocarcinomas.

This tissue-specific pattern reflects the different roles that RAS family members play in normal development and tissue homeostasis. KRAS appears to be particularly important for the epithelial tissues that give rise to the most common solid tumors.

The clinical importance of KRAS extends beyond mutation frequency. KRAS-mutant cancers tend to be particularly aggressive and resistant to many conventional therapies. For decades, this made KRAS both one of the most important oncogenes and one of the most frustrating therapeutic targets.

Therapeutic Breakthrough: The KRAS G12C Revolution

The story of KRAS as an "undruggable" target is one of the most dramatic in modern cancer medicine. For over 30 years, KRAS was considered impossible to target directly due to its high affinity for GTP and lack of obvious small molecule binding sites.

The breakthrough came with the recognition that different KRAS mutations might create different vulnerabilities. The G12C mutation, found in about 13% of lung adenocarcinomas, creates a unique cysteine residue that can be targeted by covalent inhibitors.

Sotorasib and adagrasib represent the first successful direct KRAS inhibitors. These drugs work by covalently binding to the cysteine residue created by the G12C mutation, locking KRAS in its inactive GDP-bound state. The clinical results have been remarkable, with significant response rates in previously untreatable KRAS G12C-mutant lung cancers.

This success has revitalized efforts to target other KRAS mutations and has validated the concept that even supposedly "undruggable" proteins can be targeted with sufficient ingenuity and persistence.

MYC: The Transcriptional Master Regulator

Now let's turn our attention to MYC, an oncogene family that operates through fundamentally different mechanisms than RAS. While RAS functions as a signal transduction switch, MYC acts as a transcriptional regulator that directly controls gene expression programs.

The MYC family consists of three members: c-MYC, N-MYC, and L-MYC. Of these, c-MYC is the most broadly important and the most frequently altered in human cancers. MYC represents one of the most potent oncogenes known, capable of transforming cells and driving aggressive tumor growth.

Transcriptional Regulation: The 15% Rule

What makes MYC so powerful is the sheer scope of its transcriptional influence. MYC regulates approximately 15% of all human genes, making it one of the most influential transcription factors in the human genome. This broad regulatory control explains why MYC dysregulation has such profound effects on cellular behavior.

MYC functions as part of a heterodimeric transcription factor complex with MAX (MYC-associated factor X). The MYC-MAX heterodimer binds to specific DNA sequences called E-boxes, particularly those with the core sequence CACGTG.

When MYC levels are low, as in quiescent cells, MAX forms homodimers or heterodimers with other proteins like MAD, which actively repress transcription. When MYC levels rise, as in response to growth signals, MYC-MAX heterodimers predominate and activate transcription of MYC target genes.

This creates a transcriptional switch where MYC levels determine whether target genes are repressed or activated. The switch is remarkably sensitive—relatively small changes in MYC levels can have dramatic effects on gene expression patterns.

Cellular Functions: Orchestrating the Growth Program

MYC's target genes collectively orchestrate a comprehensive cellular program focused on rapid growth and proliferation. This program touches virtually every aspect of cellular metabolism and behavior.

Cell Cycle Progression

MYC activates numerous genes involved in cell cycle progression, including cyclins D1, D2, and E, as well as cyclin-dependent kinases CDK2 and CDK4. MYC also represses genes encoding cyclin-dependent kinase inhibitors like p21 and p27.

The net effect is to accelerate cell cycle progression and reduce the cell's responsiveness to growth-inhibitory signals. MYC-overexpressing cells cycle faster and are less likely to exit the cell cycle in response to adverse conditions.

Metabolic Reprogramming

Perhaps MYC's most important function in cancer is orchestrating metabolic reprogramming. MYC activates genes involved in glucose uptake and glycolysis, including glucose transporters and virtually every glycolytic enzyme.

MYC also drives glutamine metabolism by activating genes involved in glutamine uptake and conversion to alpha-ketoglutarate. This is crucial because rapidly proliferating cells need glutamine as both a nitrogen source for amino acid synthesis and a carbon source for biosynthetic pathways.

Additionally, MYC activates the entire nucleotide biosynthesis pathway, ensuring that rapidly dividing cells have adequate supplies of DNA and RNA precursors.

Protein Synthesis

MYC massively upregulates the protein synthesis machinery. It activates genes encoding ribosomal proteins, ribosomal RNA processing factors, and translation initiation factors. This makes biological sense because rapidly proliferating cells need to produce enormous amounts of protein to support their growth and division.

Apoptosis Sensitization

Paradoxically, while MYC promotes growth and proliferation, it also sensitizes cells to apoptosis. This represents an important tumor suppressor mechanism—MYC activation in the absence of appropriate survival signals triggers cell death.

This dual function of MYC creates an important safeguard against malignant transformation. For cells to become cancerous following MYC activation, they must acquire additional alterations that provide survival signals or block apoptotic pathways.

Oncogenic Mechanisms: How MYC Goes Wrong

Unlike RAS, which is typically activated by point mutations, MYC becomes oncogenic through several different mechanisms that all result in increased MYC protein levels or stability.

Gene Amplification

One of the most common mechanisms of MYC activation is gene amplification, where tumor cells contain multiple copies of MYC genes. This leads to proportionally increased MYC expression and protein levels.

MYC amplification is particularly common in certain cancer types. N-MYC amplification is a hallmark of aggressive neuroblastomas, while c-MYC amplification is found in various solid tumors and hematological malignancies.

Chromosomal Translocations

Chromosomal translocations can place MYC genes under the control of highly active promoters, leading to inappropriate expression. The classic example is Burkitt lymphoma, where a t(8;14) translocation places c-MYC under the control of immunoglobulin heavy chain regulatory sequences.

This leads to massive MYC overexpression in B cells, driving the rapid proliferation characteristic of Burkitt lymphoma. Similar translocations involving MYC are found in other B-cell malignancies.

Post-translational Stabilization

MYC protein is normally very unstable, with a half-life of only about 30 minutes. This rapid turnover allows cells to quickly adjust MYC levels in response to changing conditions.

Various oncogenic signals can stabilize MYC protein, allowing it to accumulate to abnormally high levels. This stabilization can occur through phosphorylation events that block MYC degradation or through the loss of proteins that normally target MYC for destruction.

RAS-MYC Cooperation: A Powerful Partnership

One of the most important concepts in cancer biology is oncogene cooperation—the idea that multiple oncogenes often work together to drive malignant transformation. The relationship between RAS and MYC provides a perfect example of this cooperation.

RAS signaling directly affects MYC through multiple mechanisms. The ERK kinases activated by RAS can phosphorylate MYC at serine 62, stabilizing the protein and enhancing its transcriptional activity. The PI3K-AKT pathway activated by RAS inhibits GSK-3 β , a kinase that normally phosphorylates MYC at threonine 58 and targets it for degradation.

This creates a powerful positive feedback loop where RAS activation leads to MYC stabilization and accumulation. The accumulated MYC then drives expression of genes that further enhance cell proliferation and survival.

This cooperation helps explain why RAS and MYC alterations are frequently found together in human cancers. The combination is particularly potent because RAS provides the signaling machinery for growth and survival, while MYC provides the transcriptional program to execute rapid proliferation.

Metabolic Reprogramming: The MYC Metabolic Network

One of MYC's most important functions in cancer is coordinating metabolic reprogramming. This isn't simply about generating energy—it's about rewiring cellular metabolism to support the unique demands of rapid, uncontrolled proliferation.

Enhanced Glycolysis: The Warburg Effect

MYC is one of the key drivers of the Warburg effect—the tendency of cancer cells to preferentially use glycolysis for energy production even in the presence of oxygen. MYC activates genes encoding glucose transporters (GLUT1), hexokinase 2, and most other glycolytic enzymes.

While glycolysis is less efficient than oxidative phosphorylation for ATP production, it provides several advantages for rapidly proliferating cells. Glycolysis generates biosynthetic precursors, including ribose sugars for nucleotide synthesis and glycerol for lipid synthesis.

Glutaminolysis: The Second Fuel

MYC also drives glutamine metabolism by activating genes involved in glutamine uptake and catabolism. Glutamine serves multiple functions in rapidly proliferating cells—it provides nitrogen for amino acid and nucleotide synthesis, carbon for the TCA cycle, and reducing equivalents for biosynthetic reactions.

In many cancers, glutamine becomes as important as glucose for supporting cell growth. This glutamine addiction has become an attractive target for cancer therapy.

Nucleotide Biosynthesis: Building Blocks for Proliferation

Perhaps most importantly for cancer, MYC activates the entire nucleotide biosynthesis pathway. Rapidly dividing cells need massive amounts of DNA and RNA precursors, and MYC ensures these are available by driving expression of enzymes in both purine and pyrimidine synthesis pathways.

This metabolic reprogramming represents a clear example of how oncogenes don't just drive proliferation—they anticipate and provide for the metabolic demands that rapid proliferation creates.

Therapeutic Challenges: The "Undruggable" Transcription Factor

MYC has long been considered even more challenging to target than RAS. As a transcription factor, MYC lacks the enzymatic active sites that provide obvious targets for small molecule inhibitors. Its interactions with DNA and protein partners involve large surface areas that are difficult to disrupt with conventional drugs.

Several strategies are being pursued to target MYC:

Targeting MYC Expression

Some approaches aim to reduce MYC expression using antisense oligonucleotides or small interfering RNAs. While challenging to deliver effectively, these approaches have shown promise in preclinical models.

Disrupting Protein-Protein Interactions

MYC requires MAX as a partner to bind DNA effectively. Small molecules that disrupt the MYC-MAX interaction could potentially inhibit MYC function. This represents a challenging but potentially feasible approach to MYC inhibition.

Exploiting MYC Dependencies

MYC activation creates specific cellular dependencies that can be exploited therapeutically. For example, MYC-overexpressing cells are often dependent on glutamine metabolism, making them vulnerable to glutaminase inhibitors.

Targeting MYC-Driven Metabolic Pathways

Since MYC drives specific metabolic programs, inhibiting these pathways can selectively affect MYC-overexpressing cells. This includes targeting glycolysis, glutaminolysis, or nucleotide biosynthesis.

Integration: RAS and MYC in Cancer Biology

As we conclude our exploration of these two oncogene families, it's important to consider how they fit into the broader framework of cancer biology and how they contribute to the hallmarks of cancer.

Multiple Hallmark Contributions

Both RAS and MYC contribute to multiple cancer hallmarks, but in complementary ways:

Sustaining Proliferative Signaling: RAS provides the signaling machinery while MYC provides the transcriptional program.

Evading Growth Suppressors: Both oncogenes help cells ignore growth-inhibitory signals through different mechanisms.

Reprogramming Energy Metabolism: MYC is particularly important here, orchestrating the metabolic changes needed for rapid proliferation.

Enabling Replicative Immortality: Both contribute to overcoming senescence and enabling unlimited proliferative potential.

Therapeutic Implications

The study of RAS and MYC has had profound implications for cancer therapy:

Validation of Oncogene Targeting: The success of KRAS G12C inhibitors has validated the concept of targeting oncogenes directly.

Importance of Mutation-Specific Approaches: Different mutations may require different therapeutic strategies.

Combination Therapy Rationale: The cooperation between RAS and MYC suggests that combination approaches targeting multiple pathways may be necessary.

Metabolic Targeting: Understanding MYC-driven metabolic reprogramming has opened new avenues for cancer therapy.

Future Directions

Several trends are likely to shape the future of RAS and MYC research:

Precision Medicine: Developing therapies tailored to specific oncogene variants and combinations.

Resistance Mechanisms: Understanding how cancers develop resistance to oncogene-targeted therapies.

Combination Strategies: Optimal ways to combine oncogene-targeted therapies with other treatment modalities.

Biomarker Development: Better ways to identify patients who will benefit from specific oncogene-targeted therapies.

Conclusion: The Oncogenic Partnership

RAS and MYC represent two of the most important and well-studied oncogenes in cancer biology. While they operate through different mechanisms—RAS as a signal transduction switch and MYC as a transcriptional regulator—they often cooperate to drive malignant transformation.

Understanding these oncogenes has been fundamental to our comprehension of cancer biology and has opened important therapeutic opportunities. The recent success with KRAS inhibitors demonstrates that even challenging targets can eventually be drugged with sufficient effort and innovation.

As we continue our exploration of cancer biology, we'll see how oncogene activation works together with tumor suppressor loss, genomic instability, and other mechanisms to drive the complex process of malignant transformation.

The principles we've learned from studying RAS and MYC—the importance of signal transduction, transcriptional control, metabolic reprogramming, and oncogene cooperation—continue to guide modern cancer research and drug development.

Next time, we'll explore the other side of this molecular equation: tumor suppressor genes and how their loss contributes to cancer development. We'll see how the balance between oncogene

activation and tumor suppressor loss ultimately determines whether cells remain normal or become malignant.

Thank you for your attention, and I look forward to our discussion.